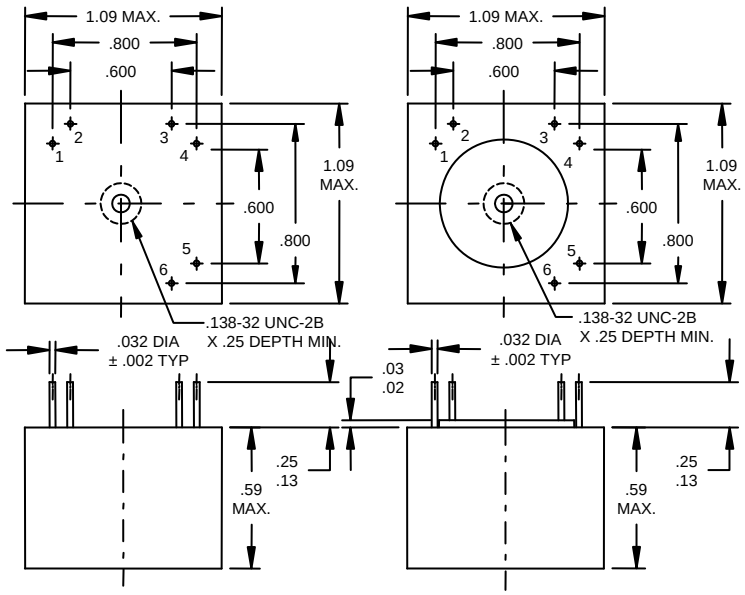
INCH-POUND
MIL-PRF-27/38C
25 September 2009
SUPERSEDING
MIL-PRF-27/38B
7 February 1984

PERFORMANCE SPECIFICATION SHEET

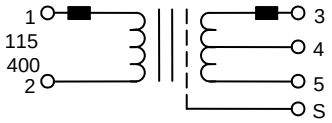
TRANSFORMERS, POWER, 1.5 VOLT AMPERES, 400 HZ, ELECTROSTATIC SHIELDED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformer described herein shall consist of this specification sheet and MIL-PRF-27.



CONFIGURATION A CONFIGURATION B



"X" VOLTAGE CT

"X" mA OR A

MAX W.V. , "X"V"
MAX ALT 70 .000

(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAM AND MARKING

Inches	mm
.002	0.05
.02	0.5
.03	0.8
.032	.81
.13	3.31
.138	3.51
.25	6.4
.59	15.0
.600	15.24
.75	19.1
.800	20.32
1.09	27.2

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.005 (0.13 mm).
4. Marking shall be on the side of the case.
5. A white dot shall be marked on the side opposite the terminals, over terminal 1, in addition to the terminal marking shown.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configurations

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REQUIREMENTS: (When numbers in parentheses, such as (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 1.1 ounces, maximum.

Duty cycle: Continuous.

Case: Plastic encapsulated.

Material: Plastic.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55˚C to +130˚C.

Terminals: Printed circuit pins - tinned brass.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics:

No load: With 115 volts rms, 400 Hz across (1-2): Current in (1-2) shall not exceed 3 mA rms.

Rated load: With 115 volts, 400 Hz across (1-2) and a restrictive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 70 percent minimum at rated load.

Electrostatic shielding: The voltage ratio shall be a minimum of 5 to 1 at 20 kHz.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{\text{Voltage (no Load) - Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$

shall not exceed 18 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 17°C, maximum.

Vibration: MIL-STD-202, method 204, test condition D.

Marking location: See figure 1.

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VERIFICATION

Extent of qualification

Qualification testing and approval to M27/37-44 A or B, M27/42-44 A or B, and M27/46-44 A or B shall be sufficient to grant qualification approval to MIL-PRF-27/37 through MIL-PRF-27/47 inclusive, all parts.

Qualification testing and approval to M27/38-44 shall be sufficient to grant qualification approval to M27/38-01 through M27/38-44.

Part or Identification Number (PIN): M27/38-01 A or B.

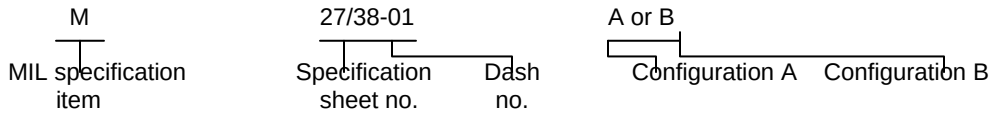

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TABLE I. Electrical ratings.

Dash No.	Type designation	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms) (MAX)
01 A or B	444	6	0.250	325
02 A or B	445	6.3	0.238	325
03 A or B	446	7	0.214	325
04 A or B	447	7.5	0.200	325
05 A or B	448	8	0.188	325
06 A or B	449	9	0.167	325
07 A or B	450	10	0.150	325
08 A or B	451	11	0.136	325
09 A or B	452	12	0.125	325
10 A or B	453	12.6	0.119	325
11 A or B	454	13	0.115	325
12 A or B	455	15	0.100	325
13 A or B	456	16	0.094	325
14 A or B	457	18	0.083	325
15 A or B	458	20	0.075	325
16 A or B	459	22	0.068	325
17 A or B	460	24	0.063	325
18 A or B	461	26	0.058	325
19 A or B	462	28	0.053	325
20 A or B	463	30	0.050	325
21 A or B	464	33	0.045	325
22 A or B	465	36	0.042	325
23 A or B	466	39	0.038	325
24 A or B	467	43	0.035	325
25 A or B	468	47	0.032	325
26 A or B	469	51	0.029	325
27 A or B	470	56	0.027	325
28 A or B	471	62	0.024	325
29 A or B	472	68	0.022	325
30 A or B	473	75	0.020	325
31 A or B	474	82	0.018	325
32 A or B	475	91	0.016	325
33 A or B	476	100	0.015	325
34 A or B	477	110	0.014	325
35 A or B	478	115	0.013	325
36 A or B	479	120	0.0125	325
37 A or B	480	130	0.0115	325
38 A or B	481	150	0.010	325
39 A or B	482	180	0.0085	325
40 A or B	483	200	0.0075	440
41 A or B	484	220	0.007	440
42 A or B	485	240	0.006	440
43 A or B	486	270	0.0055	580
44 A or B	487	300	0.005	580

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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-PRF-27/37	MIL-PRF-27/39	MIL-PRF-27/40	MIL-PRF-27/41
MIL-PRF-27/42	MIL-PRF-27/43	MIL-PRF-27/44	MIL-PRF-27/45
MIL-PRF-27/46	MIL-PRF-27/47	MIL-STD-202.	

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2009-037)

Review activities:
Army - CR4, MI
Navy - AS, MC, OS
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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